

KVCO Corner Characterization Flow

1. Testbench: Tese_Varactor_characterization_tb1

Corners: TT25, SS125, FF-40, FF125, SS-40

Sweep: Parametric sweep of V_{dc} from 0.2 V to 0.6 V (60 points).

1.1 Minimum Frequency ($f_{min} = 6$ GHz) Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
$C_{wParasitics}$	abs(imag(Device_current_M7) / (2 * 3.14159 * 6e9)) + 114.5f + 1000.07f + 1612.48f
$C_{wParasitics,cont}$	flatten(Cap_wParasitics_Cc, Vdc)
$\frac{dC}{dV}$	derivative(Cap_wParasitics_Cc_cont)
$V_{dc,opt}$	x_at_ymin(o(Deriv_Cap_measuredAt1Mhz_M7, analysisName="ac"), 0.1, 0.5, 1)
$V_{dc,opt,cont}$	flatten(deriv_Vdc_opt, Vdc)
Freq (F)	1 / (2 * 3.1416 * sqrt(236.20p * Cap_wParasitics_Cc))
F_{cont}	flatten(Freq, Vdc)
K_{VCO}	abs(derivative(Freq_cont))
$K_{VCO,max}$	ymin(o(KVCO, analysisName="ac"), 0, 0.5)
ΔK_{VCO}	max(KVCO_max) - min(KVCO_max)